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NATIONAL DEVELOPMENT COMPLEX
Deputy Directorate General of Quality Control
PHYSICAL LAB, Site-3A
TEST REPORT

Test Report No: TR (015)/PL/2022

Job Card No: JC (029)/PL/2022

Date: 07-03-2022

INDENTER'S DATA

Name:	Shamim Khan Asif	Designation:	FM-I
Group & Site:	PP / 3C	Directorate/ Department:	ADP
Organization:	NDC	Contact No:	3656

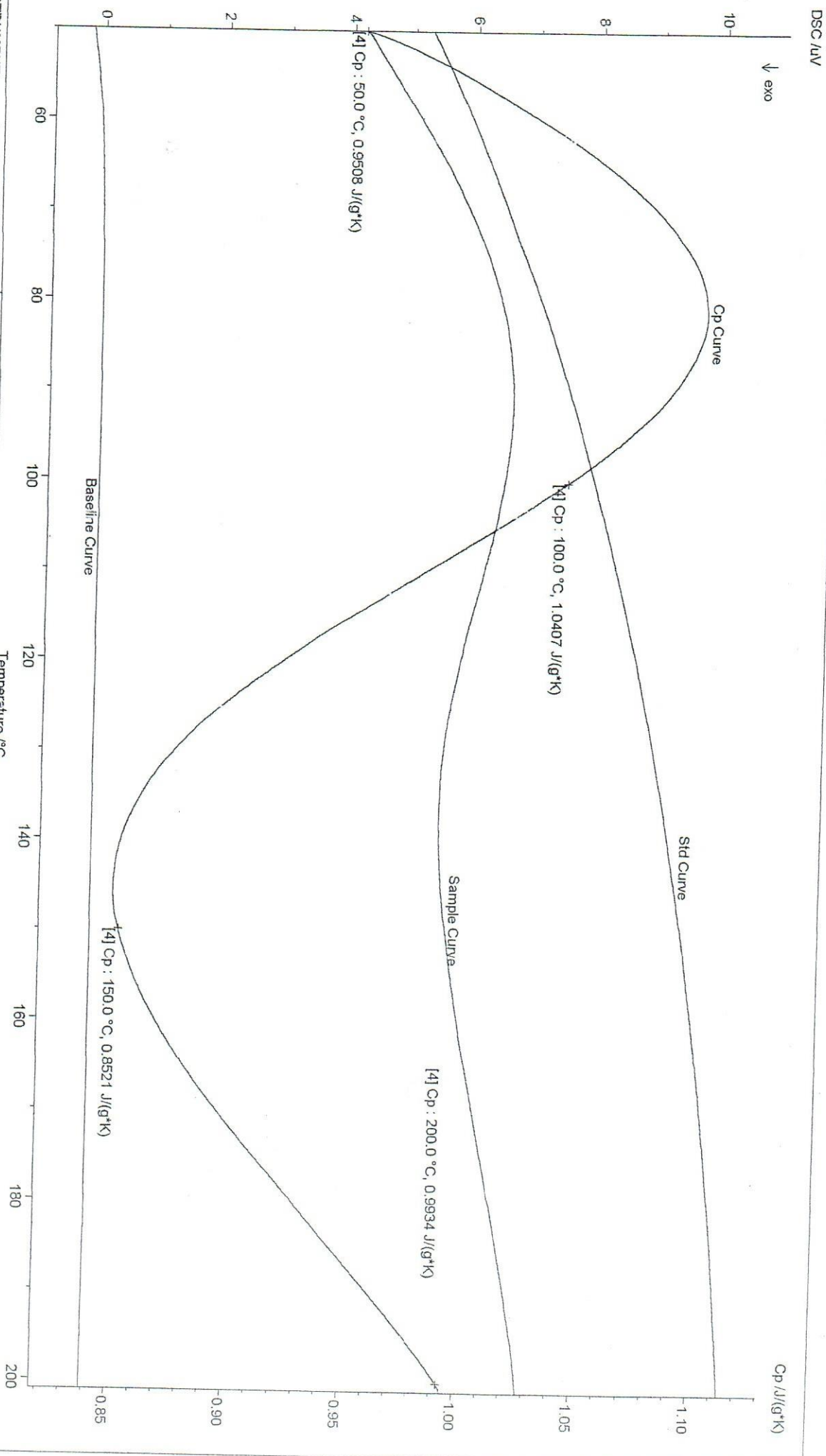
GENERAL DATA

Sample Name:	Silica Phenolic Ablative	ID No:	PP-Si-10A/1120-Cp-01~05
Lot No :	Si(820)-10A/1120-221~333	Lot Size:	7922RM
Shape:	Powder	Chemical Composition:	Silica Phenol
Project:	All Systems	Quantity:	05
Sample Receiving Date:	07-03-2022	Test Date:	07-03-2022

TEST DATA

Test Method Used	PL/SOP/06	Qualification Criteria	NDC-Abl/QC/Si-C Ablatives/02 Rev : 01
Instrument Type	DSC-404C	Testing Temperature (°C)	RT to 200°C
Temperature(°C)	21.7 ±1	Relative Humidity (%)	57 ±2
Test Type	Sample ID	Test Results	
Specific Heat Capacity (Cp)	PP-Si-10A/1120-Cp-01	0.9508 J/gK at 50.0°C	
		1.0407 J/gK at 100.0°C	
		0.8521 J/gK at 150.0°C	
		0.9934 J/gK at 200.0°C	
	PP-Si-10A/1120-Cp-02	0.8432 J/gK at 50.0°C	
		1.0434 J/gK at 100.0°C	
		0.8657 J/gK at 150.0°C	
		1.0015 J/gK at 200.0°C	
	PP-Si-10A/1120-Cp-03	0.8325 J/gK at 50.0°C	
		1.0680 J/gK at 100.0°C	
		0.9274 J/gK at 150.0°C	
		1.0538 J/gK at 200.0°C	

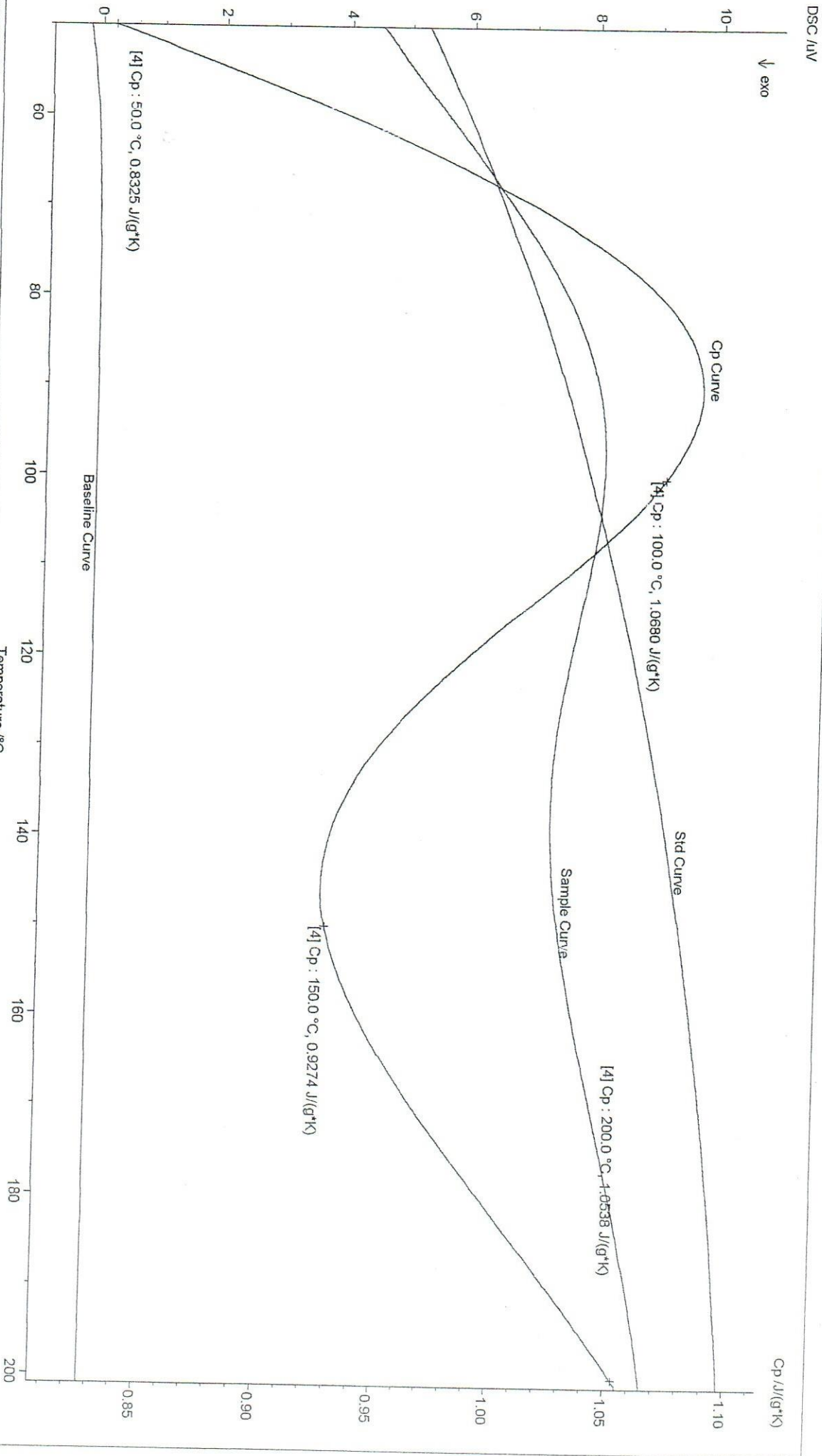




INSTRUMENT: NETZSCH DSC 404C
PROJECT: QCD
IDENTITY: PP-Si-10A/120-Cp-01
LABORATORY: Physical lab
OPERATOR: Imran

[2]SAMPLE:	Sample name	mass
[1]STANDARD FILE:	04.02.22	28.960mg
CP-STANDARD TABLE:	SAP4222.DC	41.300mg
[3]BASELINE:	NEU NBS	
[4]CP:	BLO40222.BDC	
MATERIAL:	04.02.22	
	PP-SI-1.DDC	
	Phenolic Ablative	28.960mg

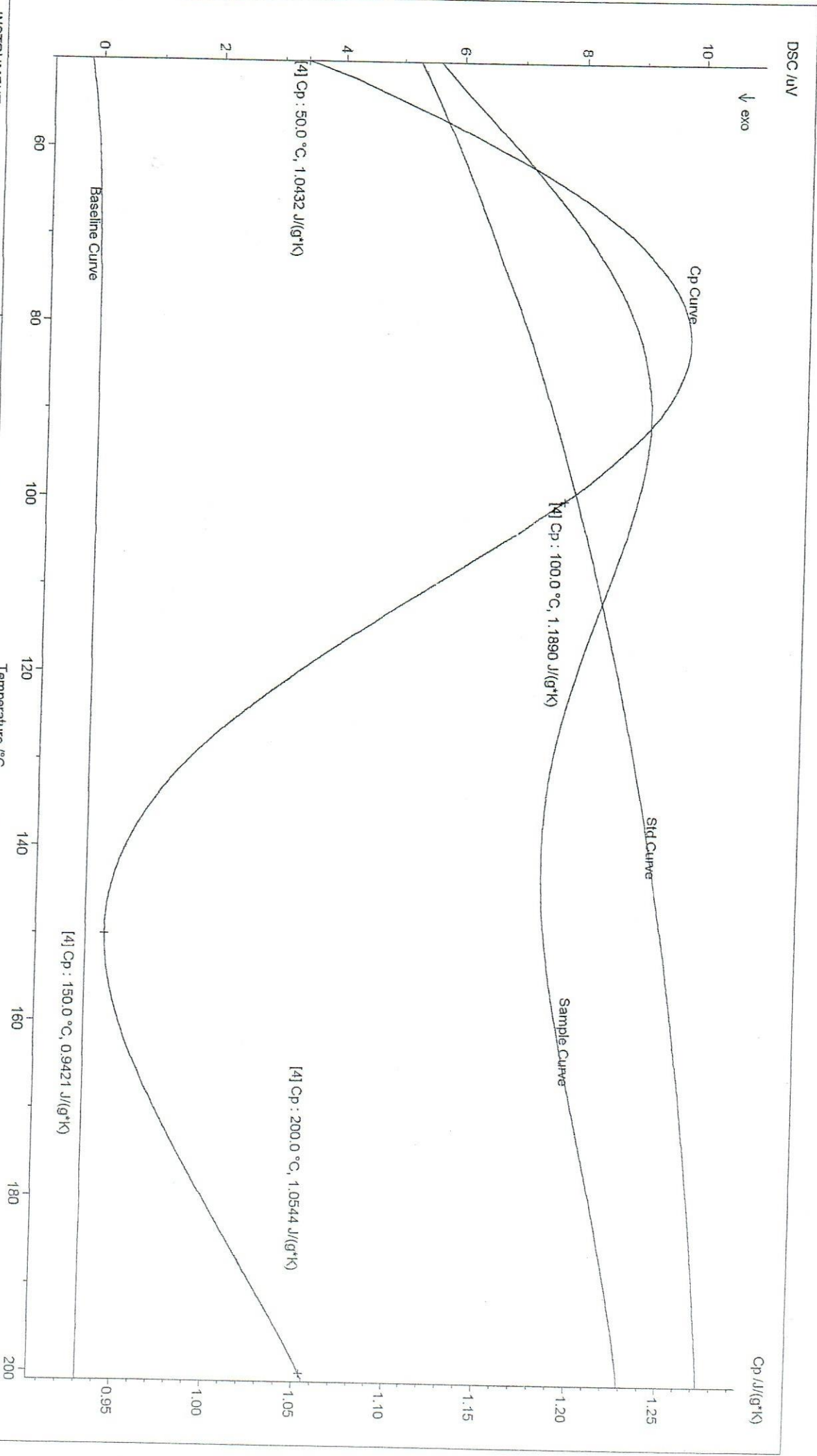
SEGMENTS:	1/ 1
SAMPLE CAR./TC:	DSC(TG) HIGH RG 2/S
CRUCIBLE:	DSC(TG) pan Pt-Rh
HEAT. RATE/(K/min):	20.0
ATMOSPHERE:	N2/70
CORRM.RANGE:	8800/500



INSTRUMENT: NETZSCH DSC 404C
PROJECT: QCD
IDENTITY: PP-Si-10A/1120-Cp-03
LABORATORY: Physical lab
OPERATOR: Imran

File	Date	Sample name	mass
[2]SAMPLE: PP-Si-3.DDC	04.02.22	Silica Phenolic Abative	35.050mg
[1]STANDARD-FILE: SAP4222.DDC	04.02.22	Sapphire	41.300mg
Co-STANDARD TABLE:			
[3]BASELINE: BLO40222.DDC	04.02.22	NBS-Standardtabelle	
[4]Cp: PP-Si-3.DDC			35.050mg
MATERIAL: a Phenolic Abative			

SEGMENTS: 1/1
SAMPLE CAR./TC: DSC(TG) HIGH RG 2/S
CRUCIBLE: DSC/TG pan Pt-Rh
HEAT. RATE /(K/min): 20.0
ATMOSPHERE: N2/70
CORR/M.RANGE: 8800/500



INSTRUMENT: NETZSCH DSC 404C
PROJECT: QCD
IDENTITY: PP-SI-10A/1120-Cp-05
LABORATORY: Physical lab
OPERATOR: Imran

File	Date	Sample name	mass
[2]SAMPLE: PP-SI-5.DDC	04.02.22	Silica Phenolic Ablative	34.700mg
[1]STANDARD-FILE: SAP4222.DDC	04.02.22	Sapphire	41.300mg
Cp-STANDARD TABLE: NEU_NBS		NBS-Standardtabelle	
[3]BASELINE: BL040222.BDC	04.02.22		
[4]Cp: PP-SI-5.DDC			34.700mg
MATERIAL: Silica Phenolic Ablative			

SEGMENTS: 1/1
SAMPLE CAR./TC: DSC(TG) HIGH RG 2/S
CRUCIBLE: DSC/TG pan Pt-Rh
HEAT. RATE. / (K/min): 20.0
ATMOSPHERE: N2/70
CORRM. RANGE: 8800/500